

Nickolas Ellison

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Education:

Harvard University, Cambridge, MA

A.B. Mathematics and Physics, with concurrent Physics Masters, A.M. GPA: 3.902 Expected May 2027

Coastal Bend College, Beeville, TX

Associate, General Sciences. GPA: 4.0. Highest Honors

Graduation May 2023

Honors and Grants:

JSSURP (Japan Summer Science Undergraduate Research Program) Grant, Harvard (2025)

Phi Theta Kappa Honor Society Scholarship, Coastal Bend College (2023)

Research Experiences:

Norman Yao Lab, Harvard, Cambridge MA

Undergraduate Researcher, September 2025 - Present

NV Diamond Centers - The general research direction of this lab is on quantum dynamics, quantum matter, and quantum information, and the specific area I am focusing on is quantum sensing using diamonds with NV (nitrogen-vacancy) centers. I worked with a post-doc on synchronizing a camera system to a laser system for measurements of materials that are of interest, after applying various laser sequences to the material.

Ultracold Atoms - I am currently within the same lab, but on a laser stabilization project, where I use a stable laser and a scanning cavity with a feedback loop to stabilize the unstable laser. Involves building the software for peak detection, the feedback loop, and signal generation/analysis. On the hardware side, it involves building optical set ups using lasers, fiber optics, beam splitters, mirrors, lenses, and cavities. Additionally it requires electronics and circuit design, for building amplifiers to be used with an FPGA (red pitaya).

RIKEN Center for Brain Science, Wako Japan

RIKEN CBS Summer Research Internship Participant, June - August 2025

Advisors: Dr. Takuya Isomura and Dr. Richard Servajean

Conducted research on the Free Energy Principle in the context of theoretical neuroscience. Focused specifically on extending the normal set up (internal states, external states, separated by a boundary of sensory and active states) to hierarchical models. Extended the mathematical formalism to these models as well as conducted simulations. Concluded with a presentation of my research.

MIT 3D Optics Group, Cambridge MA

Undergraduate Researcher, February 2025 - May 2025

Advisor: Dr. George Barbastathis

Studied a toy model for solutions containing two different types of micro particles. Mainly focused on extending Professor Barbastathis initial set up to make the model more complex and realistic. Generated and analyzed speckle images as well as an autocorrelator for these simulations. Then worked on fitting various models to the autocorrelator.

Presentations:

Harvard College, PRISE Summer Research Program, Final Presentations, Cambridge, MA, USA, August 2026:

“Frequency Stabilization Cavity Transfer” based on summer research I conducted.

RIKEN Center for Brain Science Summer Internship Final Presentations, Wako Japan, August 2025, "Hierarchical Extension of the Free Energy Principle”, based on the summer research I conducted.

RIKEN Center for Brain Science Summer Lecture Course Poster Presentations, Wako, Japan, July 2025 “Two Particle Dynamics”, based on my prior research with Professor Barbastathis.

Teaching Experiences:

Harvard College, Cambridge MA

Math 113 (Complex Analysis) Course Assistant, Jan. 2026 - May 2026.

Host Office Hours, answer questions, grade homework, and host review sections.

Math 122 (Group Theory) Course Assistant, August 2025 - December 2025.

Host Office Hours, answer questions, grade homework, write solutions, latex lectures, and host review sections.

Physics 143A (Quantum Mechanics 1) Course Assistant, January 2025 - May 2025.

Host Office Hours, answer questions, grade homework.

Math 136 (Differential Geometry) Course Assistant, Sept 2024 - December 2024.

Host Office Hours, answer questions, grade homework.

Sabine High School, Gladewater TX

Math Camp Teacher, July 2023 - July 2023 and July 2024 - July 2024

Teach high school students, organize material, host sections.

Technical Skills:

Python, MatLab, Latex, C, Java, JavaScript, html, SQL, circuit design/building, laser and optics engineering.